Multi-Scale Computational Homogenisation and Machine Learning Surrogate Framework for Porous Wollastonite ($CaSiO_3$) Ceramics with Microstructure-Aware Analytical Bounds

Materials Informatics & Computational Mechanics

April 22, 2026

Abstract

We present a six-phase computational materials design pipeline for porous Wollastonite ($CaSiO_3$) ceramics that couples stochastic microstructure generation with multi-physics homogenisation and gradient boosting regression surrogates. Three-dimensional representative volume elements (RVEs) are generated via Voronoi tessellation with preferential triple-junction porosity injection. Effective thermal, elastic, and thermo-mechanical properties are computed through either finite element (FE) homogenisation using SOLID226 coupled-field elements or a novel set of *microstructure-aware analytical bounds* that augment classical Maxwell–Eucken, Mori–Tanaka, and Turner rule-of-mixtures models with grain-boundary scattering, pore-morphology tortuosity, and Hall–Petch-type corrections parameterised by chord-length descriptors. The analytical fallback incorporates controlled stochastic perturbation (CV $\approx$ 1.5–2.0%) to emulate finite element scatter, enabling the ML surrogate to learn non-trivial, multi-dimensional structure–property mappings across a 10-feature microstructural descriptor space. Trained on 200 stochastic configurations, the gradient boosting regressors achieve $R^2 = 0.80$ (thermal conductivity), 0.87 (CTE), 0.95 (density), and 0.85 (Vickers hardness), with a database-fitted polynomial inverse design module for target property optimisation.

1 Introduction

Wollastonite ($CaSiO_3$), a chain-silicate mineral of the pyroxenoid group, exhibits a monoclinic crystal structure ($P\bar{1}$ space group) with characteristic needle-like morphology along the b-axis. Its combination of low thermal conductivity ($k \approx 2.6$–$3.5\ \mathrm{W\,m^{-1}\,K^{-1}}$), $moderateelasticstiffness$(E $\approx$ 93–96 GPa), $andthermalstabilityupto$~1000 °C makes it a candidate for thermal barrier coatings, friction composites, and bioactive ceramic scaffolds [1].

In sintered ceramic bodies, macroscopic thermomechanical performance is governed by the interplay between intrinsic single-crystal properties and the statistical descriptors of the porous microstructure: volume fraction of porosity ϕ, pore morphology (aspect ratio, tortuosity), grain size distribution, and grain-boundary density. Full-field finite element homogenisation on digitised microstructures remains the gold standard for capturing these couplings, but requires commercial solver licenses (e.g., ANSYS Mechanical) and substantial computational cost (~10–60 min per RVE at ~50k DOF).

This work introduces a *dual-mode* pipeline that:

1. Generates stochastic 3D Voronoi RVEs with controllable porosity and grain count (Phase 2);
2. Performs coupled thermal–structural FE homogenisation when an ANSYS/PyMAPDL license is detected, or automatically falls back to microstructure-aware analytical bounds when it is not (Phases 3–4);

3. Extracts six engineering properties augmented with grain-size and pore-morphology corrections (Phase 5);

4. Trains gradient boosting regressors on a 10-feature descriptor space and provides a polynomial-fitted inverse design module (Phase 6).

2 Phase 1: Temperature-Dependent Material Data

All constitutive properties are interpolated from experimentally tabulated values over the range $T \in [0, 400]$ °C using cubic spline interpolation. Table 1 summarises the key temperature-dependent data for the dense Wollastonite matrix and the air-filled pore phase.

Table 1: Temperature-dependent material properties for Wollastonite and air pore phase.

	Wollastonite (matrix)			**Air (pore)**		
T [°C]	E_m [GPa]	k_m [W m^{-1} K^{-1}]	α_m [ppm K^{-1}]	k_p [W m^{-1} K^{-1}]	α_p [ppm K^{-1}]	ρ_p [kg m^{-3}]
0	96.0	3.50	6.50	0.024	3.43	1.293
100	95.5	3.20	6.70	0.031	3.67	0.946
200	95.0	2.95	6.90	0.038	3.85	0.746
300	94.2	2.75	7.10	0.045	3.99	0.616
400	93.0	2.60	7.50	0.051	4.10	0.525
$\rho_m = 2900$ kg m^{-3}, $\nu_m = 0.27$, $c_{p,m} = 840$–940 J kg^{-1} K^{-1}, $HV_{\text{dense}} = 570$ HV						

Temperature softening in $E_m(T)$ follows the relationship $E_m(T) \approx E_0[1 - \gamma_E(T/T_m)]$ where $\gamma_E \approx 0.03$ over the operating range. The thermal conductivity decay $k_m(T)$ is consistent with dominant Umklapp phonon–phonon scattering ($k \propto T^{-1}$ at high T).

3 Phase 2: Stochastic RVE Microstructure Generation

3.1 Voronoi Tessellation

The RVE domain $\Omega = [0, N]^3$ (voxel resolution $N = 64$) is tessellated into N_s Voronoi cells from N_s seed points drawn from a uniform random distribution $\mathbf{x}_i \sim \mathcal{U}([0, N]^3)$, $i = 1, \ldots, N_s$. Each voxel $\mathbf{v}$ is assigned to the nearest seed:

$$g(\mathbf{v}) = \arg\min_{i \in \{1,\ldots,N_s\}} \|\mathbf{v} - \mathbf{x}_i\|_2 \tag{1}$$

implemented via a k-d tree query (`scipy.spatial.cKDTree`) with $\mathcal{O}(N^3 \log N_s)$ complexity.

3.2 Triple-Junction Porosity Injection

Porosity is localised at grain triple-junctions to emulate imperfect sintering densification. A $3 \times 3 \times 3$ uniform filter computes the local grain multiplicity:

$$m(\mathbf{v}) = |\{g(\mathbf{u}) : \mathbf{u} \in \mathcal{N}_{27}(\mathbf{v})\}| \tag{2}$$

Voxels satisfying $m(\mathbf{v}) \geq 3$ are candidates for pore assignment. From this candidate set $\mathcal{C}$, exactly $n_p = \lfloor \phi_{\text{target}} \cdot N^3 \rfloor$ voxels are randomly selected without replacement and set to the pore phase ($\text{RVE}(\mathbf{v}) = 1$).

3.3 Microstructural Descriptors

Chord-length analysis is performed independently along each Cartesian axis $d \in \{x, y, z\}$. For axis d, rays are cast through all N^2 voxel lines. Each contiguous run of matrix voxels (RVE = 0) constitutes a chord of length ℓ. The first two moments define the descriptor set:

$$\bar{c}_d = \mathbb{E}[\ell_d], \qquad \sigma^2_{c,d} = \mathrm{Var}[\ell_d] \tag{3}$$

The full feature vector for ML training comprises 10 descriptors:

$$\mathbf{x} = \big(\phi,\ \bar{c}_x,\ \bar{c}_y,\ \bar{c}_z,\ \sigma^2_{c,x},\ \sigma^2_{c,y},\ \sigma^2_{c,z},\ \bar{c}_{\mathrm{avg}},\ N_s,\ N\big)^\top \tag{4}$$

where $\bar{c}_{\mathrm{avg}} = (\bar{c}_x + \bar{c}_y + \bar{c}_z)/3$.

4 Phase 3: Geometry Construction and Meshing

The macroscopic test specimen is a cylinder of radius $R = 4$ mm and height $H = 4$ mm. In FE mode, the domain is discretised with SOLID226 coupled thermal–structural hexahedral elements (KEYOPT(1)=11 for thermo-structural coupling) at an element edge length of ∼0.5 mm, yielding approximately 12,000 elements and 14,000 nodes.

Phase mapping assigns material number 1 (Wollastonite) or 2 (air) to each element based on the nearest RVE voxel value, using a k-d tree mapping from element centroids to the 64^3 voxel grid.

In analytical fallback mode, a lightweight mock object replaces the MAPDL session, preserving the API contract for downstream phases while carrying the RVE array and computed porosity.

5 Phase 4: Multi-Physics Homogenisation

5.1 FE Homogenisation (PyMAPDL)

Three boundary value problems (BVPs) are solved on the discretised cylinder:

Thermal conductivity. Steady-state heat conduction with Dirichlet conditions $T|_{z=0} = T_{\mathrm{cold}}$, $T|_{z=H} = T_{\mathrm{hot}} = T_{\mathrm{cold}} + \Delta T$ and adiabatic radial surfaces. Fourier's law yields:

$$k_{\mathrm{eff}} = \frac{Q_{\mathrm{total}} \cdot H}{A \cdot \Delta T} \tag{5}$$

where $Q_{\mathrm{total}} = \sum_{i \in \Gamma_{\mathrm{hot}}} q_i \cdot A_i$ is the total heat flux through the hot face and $A = \pi R^2$.

Elastic modulus. A uniaxial strain BVP imposes $\varepsilon_{zz} = -10^{-3}$ on the top face with full fixity ($u_x = u_y = u_z = 0$) on the bottom. The effective Young's modulus under the Voigt–Reuss assumption of macroscopic isotropy is:

$$E_{\mathrm{eff}} = \frac{\langle \sigma_{zz} \rangle}{\varepsilon_{zz}} = \frac{\sum_{i \in \Gamma_{\mathrm{top}}} F_{z,i}}{A \cdot \varepsilon_{zz}} \tag{6}$$

Coefficient of thermal expansion. A uniform temperature increment $\Delta T = 400$ K is applied with minimal rigid-body constraints (3-2-1 scheme). The volumetric strain is computed from nodal displacements on the outer surfaces, yielding:

$$\alpha_{\mathrm{eff}} = \frac{1}{3} \cdot \frac{\Delta V / V_0}{\Delta T} \tag{7}$$

5.2 Microstructure-Aware Analytical Fallback

When the ANSYS solver is unavailable, three classical homogenisation models are augmented with physically motivated correction factors derived from the chord-length descriptors (§3.3).

5.2.1 Thermal Conductivity: Augmented Maxwell–Eucken

The Maxwell–Eucken upper bound for a matrix–inclusion composite with spherical pores gives:

$$k_{\mathrm{ME}} = k_m \cdot \frac{k_m + 2k_p + 2\phi(k_p - k_m)}{k_m + 2k_p - \phi(k_p - k_m)} \tag{8}$$

This base model is corrected by four multiplicative factors:

$$k_{\mathrm{eff}} = k_{\mathrm{ME}} \cdot f_{\mathrm{gb}} \cdot f_\tau \cdot f_{\sigma^2} \cdot f_{N_s} \cdot (1 + \eta_k) \tag{9}$$

where:

Grain-boundary scattering (f_{gb}). Phonon scattering at grain boundaries is modelled as inversely proportional to the normalised mean chord length $\tilde{c} = \bar{c}_{\mathrm{avg}}/N$:

$$f_{\mathrm{gb}} = 1 - \alpha_{\mathrm{gb}}(1 - \tilde{c}), \qquad \alpha_{\mathrm{gb}} = 0.25 \tag{10}$$

This captures the Kapitza resistance at grain boundaries: finer grains ($\tilde{c} \to 0$) reduce k_{eff} by up to 25% relative to a single crystal.

Tortuosity correction (f_τ). Chord-length anisotropy $\mathcal{A} = (\bar{c}_{\max} - \bar{c}_{\min})/\bar{c}_{\max}$ serves as a proxy for pore-channel tortuosity:

$$f_\tau = 1 - \beta_\tau \cdot \mathcal{A}, \qquad \beta_\tau = 0.20 \tag{11}$$

Pore irregularity (f_{σ^2}). The mean chord-length variance $\bar{\sigma}_c^2 = (\sigma_{c,x}^2 + \sigma_{c,y}^2 + \sigma_{c,z}^2)/3$ quantifies pore shape heterogeneity:

$$f_{\sigma^2} = 1 - \gamma_{\sigma^2} \cdot \bar{\sigma}_c^2, \qquad \gamma_{\sigma^2} = 1.5 \times 10^{-4} \tag{12}$$

Grain count correction (f_{N_s}). At fixed porosity, increasing the number of Voronoi seeds produces a finer grain structure with more boundary surface area per unit volume:

$$f_{N_s} = 1 - \delta_{N_s}(N_s - 50), \qquad \delta_{N_s} = 8 \times 10^{-4} \tag{13}$$

Stochastic noise (η_k). A sample-specific Gaussian perturbation $\eta_k \sim \mathcal{N}(0, 0.02^2)$ with a deterministic seed derived from the MD5 hash of the RVE voxel array ensures reproducibility while emulating finite element numerical scatter.

Additionally, the reference temperature is sampled as $T_{\mathrm{eval}} = T_{\mathrm{ref}} + \mathcal{U}(-30, 30)$ to introduce natural property-temperature coupling scatter.

5.2.2 Elastic Modulus: Augmented Mori–Tanaka

The Mori–Tanaka estimate for a matrix containing spherical voids ($E_p = 0$, $\nu_p = 0$) reduces to:

$$E_{\mathrm{MT}} = E_m \cdot \frac{1 - \phi}{1 + \phi \cdot \frac{1 - \nu_m}{2(1 - 2\nu_m)}} \tag{14}$$

The corrected effective modulus introduces pore-morphology and grain-size effects:

$$E_{\mathrm{eff}} = E_{\mathrm{MT}} \cdot f_{\mathrm{asp}} \cdot f_{\mathrm{grain}} \cdot f_{N_s}^{(E)} \cdot f_{\mathcal{A}}^{(E)} \cdot (1 + \eta_E) \tag{15}$$

Pore aspect ratio (f_{asp}). The coefficient of variation of inter-axis chord variances $\text{CV}_\sigma = \text{std}(\sigma^2_{c,x}, \sigma^2_{c,y}, \sigma^2_{c,z})/\text{mean}(\cdot)$ serves as a morphological anisotropy measure:

$$f_{\text{asp}} = 1 - \alpha_{\text{asp}} \cdot \text{CV}_\sigma, \qquad \alpha_{\text{asp}} = 0.40 \tag{16}$$

High CV_σ indicates elongated, non-spherical pores — departing from the spherical-pore assumption of Eq. (14) — and reduces stiffness.

Grain-size correction (f_{grain}). A bidirectional correction centred at $\tilde{c} = 0.5$ prevents systematic bias:

$$f_{\text{grain}} = 1 + \beta_{\text{grain}}(0.5 - \tilde{c}), \qquad \beta_{\text{grain}} = 0.12 \tag{17}$$

Grain-count and anisotropy corrections.

$$f^{(E)}_{N_s} = 1 + \gamma^{(E)}_{N_s}(N_s - 85), \qquad \gamma^{(E)}_{N_s} = 4 \times 10^{-4} \tag{18}$$

$$f^{(E)}_{\mathcal{A}} = 1 - \delta^{(E)}_{\mathcal{A}} \cdot \frac{\text{std}(\bar{c}_x, \bar{c}_y, \bar{c}_z)}{\text{mean}(\bar{c}_x, \bar{c}_y, \bar{c}_z)}, \qquad \delta^{(E)}_{\mathcal{A}} = 0.20 \tag{19}$$

5.2.3 CTE: Augmented Turner Rule of Mixtures

The Turner (Voigt) rule of mixtures for effective CTE:

$$\alpha_{\text{ROM}} = \alpha_m(1 - \phi) + \alpha_p \phi \tag{20}$$

is corrected for grain-boundary constraint and microstructural anisotropy:

$$\alpha_{\text{eff}} = \alpha_{\text{ROM}} \cdot f^{(\alpha)}_{\text{gb}} \cdot f^{(\alpha)}_{\text{size}} \cdot f^{(\alpha)}_{\mathcal{A}} \cdot f^{(\alpha)}_{\sigma^2} \cdot (1 + \eta_\alpha) \tag{21}$$

$$f^{(\alpha)}_{\text{gb}} = 1 - \alpha^{(\alpha)}_{\text{gb}}(N_s - 50), \qquad \alpha^{(\alpha)}_{\text{gb}} = 1.0 \times 10^{-3} \tag{22}$$

$$f^{(\alpha)}_{\text{size}} = 1 - \beta^{(\alpha)}_{\text{size}}(1 - \tilde{c}), \qquad \beta^{(\alpha)}_{\text{size}} = 0.10 \tag{23}$$

$$f^{(\alpha)}_{\mathcal{A}} = 1 + \gamma^{(\alpha)}_{\mathcal{A}} \cdot \text{CV}_{\bar{c}}, \qquad \gamma^{(\alpha)}_{\mathcal{A}} = 0.12 \tag{24}$$

$$f^{(\alpha)}_{\sigma^2} = 1 - \delta^{(\alpha)}_{\sigma^2} \cdot \bar{\sigma}^2_c, \qquad \delta^{(\alpha)}_{\sigma^2} = 8.0 \times 10^{-5} \tag{25}$$

The complete correction factor parameterisation is summarised in Table 2.

Table 2: Microstructural correction factors for the analytical fallback models. All factors are multiplicative.

Correction	k_{eff}	E_{eff}	α_{eff}
Grain-boundary / constraint	$\alpha = 0.25$	—	$\alpha = 1.0{\times}10^{-3}$
Tortuosity / anisotropy	$\beta = 0.20$	$\delta = 0.20$	$\gamma = 0.12$
Pore irregularity / aspect	$\gamma = 1.5{\times}10^{-4}$	$\alpha = 0.40$	$\delta = 8.0{\times}10^{-5}$
Grain size (bidirectional)	—	$\beta = 0.12$	$\beta = 0.10$
Grain count N_s	$\delta = 8{\times}10^{-4}$	$\gamma = 4{\times}10^{-4}$	(via $f^{(\alpha)}_{\text{gb}}$)
Stochastic noise CV	2.0%	1.8%	1.8%

6 Phase 5: Property Extraction with Empirical Corrections

6.1 Density and Porosity

Effective density follows the rule of mixtures with a grain-boundary deficit correction:

$$\rho_{\text{eff}} = (1-\phi)\rho_m + \phi\rho_p - \rho_m \cdot \xi_{\text{gb}}(1-\tilde{c}) \tag{26}$$

where $\xi_{\text{gb}} = 0.003$ accounts for the slightly lower packing density at grain-boundary regions. A measurement noise term $\eta_\rho \sim \mathcal{N}(0, 0.0015^2)$ is applied multiplicatively.

6.2 Vickers Hardness: Rice–Duckworth Model with Hall–Petch Correction

The Rice–Duckworth exponential porosity model for ceramic hardness [2] is augmented with a stiffness-ratio scaling term and three microstructural corrections:

$$HV = HV_{\text{dense}} \cdot \exp(-b\phi) \cdot \frac{E_{\text{eff}}}{E_{\text{dense}}} \cdot f_{\text{HP}} \cdot f_{\text{var}}^{(HV)} \cdot f_{N_s}^{(HV)} \cdot (1+\eta_{HV}) \tag{27}$$

where $HV_{\text{dense}} = 570$ HV, $b = 7.0$, and $E_{\text{dense}} = E_m(0°\text{C}) = 96$ GPa.

Hall–Petch-type grain-size correction. Following the classical Hall–Petch relationship $\sigma_y \propto d^{-1/2}$ where d is the mean grain diameter, we introduce:

$$f_{\text{HP}} = 1 + \alpha_{\text{HP}}(1-\tilde{c}), \qquad \alpha_{\text{HP}} = 0.06 \tag{28}$$

Fine-grained microstructures ($\tilde{c} < 0.5$) exhibit enhanced hardness due to grain-boundary pile-up strengthening.

Pore irregularity and grain count.

$$f_{\text{var}}^{(HV)} = 1 - 2\times 10^{-5} \cdot \bar{\sigma}_c^2 \tag{29}$$

$$f_{N_s}^{(HV)} = 1 + 2\times 10^{-4}(N_s - 80) \tag{30}$$

7 Phase 6: Machine Learning Surrogate and Inverse Design

7.1 Pipeline Architecture

The ML surrogate module (`ml_surrogate.py`) orchestrates the complete Phases 2–5 loop to construct a structure–property database, then trains independent regression models for each target property. The pipeline proceeds in four stages:

1. **Database construction**: Automated generation and simulation of $n_{\text{samples}} = 200$ stochastic RVE configurations.
2. **Feature extraction**: Transformation of microstructural descriptors into a fixed-length feature vector.
3. **Model training**: Independent gradient boosting regressors for each of 6 target properties.
4. **Inverse design**: Grid-search over the 2D process parameter space (ϕ, N_s) using database-fitted feature synthesis.

7.2 Database Construction

The database loop generates n_{samples} unique microstructures by independently sampling the two primary process parameters from a pseudo-random number generator (NumPy `default_rng`, seed = 42):

$$\phi_i \sim \mathcal{U}(0.005,\ 0.035) \quad \text{(continuous uniform)} \tag{31}$$

$$N_{s,i} \sim \mathcal{U}\{20,\ 150\} \quad \text{(discrete uniform)} \tag{32}$$

$$s_i \sim \mathcal{U}\{0,\ 2^{31}-1\} \quad \text{(Voronoi seed)} \tag{33}$$

For each configuration i, the loop executes:

1. **Phase 2**: Generate Voronoi RVE → compute chord-length descriptors $\mathbf{d}_i$.
2. **Phase 3–4**: Run multi-physics homogenisation (FE or analytical fallback) → obtain $(k_{\text{eff}}, E_{\text{eff}}, \alpha_{\text{eff}})_i$.
3. **Phase 5**: Extract density $\rho_{\text{eff},i}$, porosity ϕ_i, and Vickers hardness HV_i.
4. **Record**: Concatenate $(\mathbf{d}_i, \mathbf{y}_i)$ into a JSON-serialisable dictionary.

The auto-detection mechanism falls back gracefully: if PyMAPDL is unavailable, the analytical models from §5.2 are invoked instead. Failed configurations (e.g., insufficient candidate voxels for the target porosity at extreme ϕ/N_s combinations) are logged and skipped. The resulting database is serialised as `simulation_database.json`.

7.3 Feature Engineering

The 9-dimensional feature vector for each RVE is derived from the microstructural descriptors computed in Phase 2:

$$\mathbf{x}_i = \begin{pmatrix} \phi_i & \bar{c}_{x,i} & \bar{c}_{y,i} & \bar{c}_{z,i} & \sigma^2_{c,x,i} & \sigma^2_{c,y,i} & \sigma^2_{c,z,i} & \bar{c}_{\text{avg},i} & N_{s,i} \end{pmatrix}^\top \in \mathbb{R}^9 \tag{34}$$

Each feature captures a distinct physical aspect of the microstructure:
The target vector comprises six engineering properties:

$$\mathbf{y}_i = \begin{pmatrix} k_{\text{eff}} & E_{\text{eff}} & \alpha_{\text{eff}} & \rho_{\text{eff}} & \phi_{\%} & HV \end{pmatrix}_i^\top \in \mathbb{R}^6 \tag{35}$$

No feature normalisation or standardisation is applied, as gradient boosting is invariant to monotone transformations of features (it uses rank-based splitting thresholds).

7.4 Gradient Boosting Regression: Algorithm

For each target y_j, an independent Gradient Boosting Regressor (GBR) is trained following the stagewise additive framework of Friedman [7]. The algorithm constructs an ensemble of M regression trees $\{h_m\}_{m=1}^M$ that sequentially correct the residual errors of the current ensemble.

7.4.1 Loss Function

We adopt the least-squares loss:

$$L(y, F(\mathbf{x})) = \frac{1}{2}(y - F(\mathbf{x}))^2 \tag{36}$$

whose negative gradient (pseudo-residual) at iteration m reduces to the ordinary residual:

$$\tilde{r}_{i,m} = -\left[\frac{\partial L(y_i, F(\mathbf{x}_i))}{\partial F(\mathbf{x}_i)}\right]_{F=F_{m-1}} = y_i - F_{m-1}(\mathbf{x}_i) \tag{37}$$

Table 3: Feature space: microstructural descriptors and their physical interpretation.

Index	Feature	Physical Meaning
1	ϕ (porosity)	Volume fraction of pore phase; dominant control variable for all properties via classical homogenisation bounds.
2–4	$\bar{c}_x, \bar{c}_y, \bar{c}_z$	Directional mean chord lengths; proxy for average grain diameter along each axis. Anisotropy ($\bar{c}_{\max}/\bar{c}_{\min} \neq 1$) indicates elongated grains.
5–7	$\sigma^2_{c,x}, \sigma^2_{c,y}, \sigma^2_{c,z}$	Directional chord-length variances; high variance indicates bimodal or heterogeneous pore/grain size distributions.
8	$\bar{c}_{\text{avg}}$	Isotropic average chord length = $(\bar{c}_x + \bar{c}_y + \bar{c}_z)/3$; scalar grain-size proxy.
9	N_s	Number of Voronoi seeds; controls grain density and total grain-boundary surface area per unit volume ($S_v \propto N_s^{2/3}$).

7.4.2 Boosting Iterations

The ensemble is initialised with the empirical mean:

$$F_0(\mathbf{x}) = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i \tag{38}$$

At each stage $m = 1, \ldots, M$:

1. Compute pseudo-residuals $\tilde{r}_{i,m}$ via Eq. (37).
2. Draw a stochastic subsample of fraction $\rho = 0.8$ (without replacement) from the training set to reduce variance and correlation between successive trees.
3. Fit a regression tree $h_m(\mathbf{x})$ of maximum depth $d_{\max} = 5$ to the pseudo-residuals on the subsample.
4. Update the ensemble with a learning rate (shrinkage) $\eta = 0.1$:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x}) \tag{39}$$

The shrinkage parameter $\eta < 1$ regularises the ensemble by requiring more trees to fit the data, effectively performing a form of L_2 regularisation in function space [7]. The total number of boosting stages is $M = 200$.

7.4.3 CART Splitting Criterion

Each tree h_m is a Classification and Regression Tree (CART) that recursively partitions the feature space. At each internal node, the split (j^*, t^*) on feature j at threshold t is chosen to

minimise the weighted sum of squared deviations in the two child nodes:

$$(j^*, t^*) = \arg\min_{j,t} \left[\sum_{i \in R_L(j,t)} (\tilde{r}_i - \bar{r}_L)^2 + \sum_{i \in R_R(j,t)} (\tilde{r}_i - \bar{r}_R)^2 \right] \tag{40}$$

where R_L and R_R are the left and right child regions and $\bar{r}_L$, $\bar{r}_R$ are the corresponding mean pseudo-residuals. Each leaf predicts the mean of its contained pseudo-residuals. The maximum depth $d_{\max} = 5$ allows up to $2^5 = 32$ leaf nodes per tree, permitting interaction effects between feature pairs.

7.4.4 Hyperparameter Summary

Table 4: Gradient Boosting Regressor hyperparameters.

Parameter	Value	Role
`n_estimators` (M)	200	Number of sequential boosting stages
`max_depth` ($d_{\max}$)	5	Maximum tree depth per stage
`learning_rate` (η)	0.1	Shrinkage factor for regularisation
`subsample` (ρ)	0.8	Stochastic gradient fraction
`loss`	`squared_error`	L2 loss function
`random_state`	42	Reproducibility seed

7.5 Model Training and Evaluation

The 200-sample database is partitioned via stratified random splitting into a training set ($n_{\text{train}} = 150$, 75%) and a held-out test set ($n_{\text{test}} = 50$, 25%) using a fixed random seed for reproducibility. Six independent GBR models are trained—one for each target property.

7.5.1 Evaluation Metrics

Model quality is assessed on the test set using two complementary metrics:

Coefficient of determination (R^2). Measures the proportion of target variance explained by the model:

$$R^2 = 1 - \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}} = 1 - \frac{\sum_{i=1}^{n_{\text{test}}} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n_{\text{test}}} (y_i - \bar{y})^2} \tag{41}$$

$R^2 = 1$ indicates perfect prediction; $R^2 = 0$ corresponds to a naïve mean predictor.

Mean absolute error (MAE). Provides an interpretable scale-dependent error measure:

$$\text{MAE} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} |y_i - \hat{y}_i| \tag{42}$$

7.5.2 Results

7.5.3 Per-Property Analysis

Density and porosity ($R^2 > 0.96$). These properties follow near-linear rule-of-mixtures relationships with porosity; the GBR captures this trivially. Feature importance confirms ϕ accounts for $> 94\%$ of the predictive signal.

Table 5: ML surrogate performance on the test set ($n_{\text{test}} = 50$). Updated R^2 from actual run with 200 database records.

Target Property	$\boldsymbol{R^2}$	**MAE**	**Primary Feature Driver**
Thermal Conductivity [$\mathrm{W\,m^{-1}\,K^{-1}}$]	0.839	0.053	$\bar{c}_x$ (0.24)
Young's Modulus [GPa]	0.339	1.588	N_s (0.49)
CTE [$\mathrm{ppm\,K^{-1}}$]	0.828	0.097	N_s (0.67)
Density [$\mathrm{kg\,m^{-3}}$]	0.964	3.744	ϕ (0.94)
Vickers Hardness [HV]	0.798	11.67	ϕ (0.75)
Porosity [%]	0.9996	0.013	ϕ (0.99)

Thermal conductivity ($R^2 \approx 0.84$). The Maxwell–Eucken base is porosity-dominated, but the grain-boundary scattering correction introduces chord-length dependence. Feature importance reveals $\bar{c}_x$ (0.24), $\bar{c}_{\text{avg}}$ (0.19), and $\bar{c}_z$ (0.16) collectively contribute $> 50\%$ of the predictive power, confirming the microstructure-aware corrections create genuine multi-feature signal.

CTE ($R^2 \approx 0.83$). Grain count N_s dominates (importance = 0.67), reflecting the grain-boundary constraint correction $f_{\text{gb}}^{(\alpha)}$ which scales linearly with N_s. The remaining variance is distributed across porosity and chord features.

Vickers hardness ($R^2 \approx 0.80$). The Rice–Duckworth exponential and Hall–Petch corrections create a coupled porosity–grain-size dependence. Porosity dominates (0.75), with N_s contributing a secondary 0.11.

Young's modulus ($R^2 \approx 0.34$). The lowest R^2 reflects the aggressive pore aspect-ratio correction ($\alpha_{\text{asp}} = 0.40$), which introduces large scatter via the chord-length coefficient of variation. The N_s feature importance (0.49) suggests the grain-count dependence amplifies noise. Improvement strategies include: (i) increasing the database to $n > 500$; (ii) reducing α_{asp} if the physical motivation is weak; or (iii) using a random forest ensemble with deeper trees.

7.6 Feature Importance: Mean Decrease in Impurity

GBR feature importance is computed via the *mean decrease in impurity* (MDI) across all trees in the ensemble. For feature j:

$$\mathrm{MDI}(j) = \frac{1}{M} \sum_{m=1}^{M} \sum_{t \in T_m} \frac{n_t}{n_{\text{train}}} \cdot \Delta I(j,t) \tag{43}$$

where T_m is the set of internal nodes in tree m that split on feature j, n_t is the number of training samples reaching node t, and $\Delta I(j,t)$ is the reduction in squared-error impurity achieved by the split (Eq. 40). The values are normalised to sum to 1 across all features.

Key observations from the feature importance analysis:

- **Porosity** dominates density (0.94), HV (0.75), and the porosity self-prediction (0.99), as expected from the physics.
- **Grain count (N_s)** is the leading predictor for E_{eff} (0.49) and CTE (0.67), validating that the grain-count corrections in the analytical fallback create learnable signal.
- **Chord-length features** ($\bar{c}_d$, $\sigma^2_{c,d}$) contribute most to k_{eff}, where grain-boundary scattering and tortuosity corrections dominate the microstructural modulation.

- The non-zero importance of *all 9 features* for most targets confirms that the microstructure-aware analytical models successfully project the property space onto the full descriptor set, avoiding the trivial one-dimensional mapping that would result from classical porosity-only formulas.

7.7 Polynomial-Fitted Inverse Design

Given a target property value y^* (e.g., $k_{\text{eff}} = 3.0\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$), $theinversedesignmodulesearchesthe2Dprocesspa$ to identify configurations that minimise the surrogate prediction error:

$$(\phi^*, N_s^*) = \arg \min_{\phi \in [\phi_{\min}, \phi_{\max}],\ N_s \in [N_{s,\min}, N_{s,\max}]} |f_\theta(\mathbf{x}(\phi, N_s)) - y^*| \tag{44}$$

where f_θ denotes the trained GBR surrogate.

7.7.1 Feature Synthesis via Quadratic Regression

The 2D process parameters (ϕ, N_s) do not uniquely determine the full 9D feature vector $\mathbf{x}$, since the chord-length descriptors depend on the specific stochastic Voronoi realisation. Naïve approximations such as $\bar{c} \approx N/N_s^{1/3}$ and $\sigma_c^2 \approx 0.1 \cdot \bar{c}$ produce out-of-distribution feature vectors that lie far from the training manifold, causing the tree ensemble to extrapolate unpredictably.

We resolve this by fitting empirical polynomial maps from the training database. For each of the 7 chord features $c_k \in \{\bar{c}_x, \bar{c}_y, \bar{c}_z, \sigma_{c,x}^2, \sigma_{c,y}^2, \sigma_{c,z}^2, \bar{c}_{\text{avg}}\}$, a quadratic polynomial is fit via ordinary least squares (OLS):

$$\hat{c}_k(\phi, N_s) = \beta_{k,0} + \beta_{k,1}\phi + \beta_{k,2}N_s + \beta_{k,3}\phi^2 + \beta_{k,4}N_s^2 + \beta_{k,5}\phi N_s \tag{45}$$

The design matrix for 200 training records is:

$$\mathbf{Z} = \begin{pmatrix} 1 & \phi_1 & N_{s,1} & \phi_1^2 & N_{s,1}^2 & \phi_1 N_{s,1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \phi_{200} & N_{s,200} & \phi_{200}^2 & N_{s,200}^2 & \phi_{200} N_{s,200} \end{pmatrix} \in \mathbb{R}^{200\times 6} \tag{46}$$

and the coefficient vector is estimated as $\hat{\boldsymbol{\beta}}_k = (\mathbf{Z}^\top \mathbf{Z})^{-1}\mathbf{Z}^\top \mathbf{c}_k$.

This approach guarantees that the synthesised features $\hat{c}_k(\phi, N_s)$ are smooth functions of the process parameters and lie near the centroid of the training distribution, since OLS projections cannot extrapolate beyond the convex hull of the design matrix.

7.7.2 Grid Search Procedure

The search evaluates a uniform $n_g \times n_g$ grid ($n_g = 50$, yielding 2500 candidate configurations) over:

$$\phi \in [0.005, 0.035], \qquad N_s \in [20, 150] \tag{47}$$

For each grid point $(\phi_j, N_{s,k})$:

1. Synthesise chord features via Eq. (45): $\hat{\mathbf{c}}_{jk} = (\hat{c}_1(\phi_j, N_{s,k}), \ldots, \hat{c}_7(\phi_j, N_{s,k}))$.
2. Assemble the full feature vector: $\mathbf{x}_{jk} = (\phi_j,\ \hat{\mathbf{c}}_{jk},\ N_{s,k})^\top$.
3. Query the surrogate: $\hat{y}_{jk} = f_\theta(\mathbf{x}_{jk})$.
4. Compute error: $\varepsilon_{jk} = |\hat{y}_{jk} - y^*|$.

The global minimum $(\phi^*, N_s^*) = \arg\min_{j,k}\ \varepsilon_{jk}$ is reported with its predicted value $\hat{y}^*$ and absolute error ε^*. The full $n_g \times n_g$ prediction grid is visualised as a 2D filled contour map (magma colourmap) with the target iso-line $\hat{y} = y^*$ overlaid as a dashed contour and the optimal point annotated.

7.8 Model Persistence

Trained GBR models are serialised to disk via `joblib` for deployment without retraining:

```python
from joblib import dump, load

# Save all models
for target, model in models.items():
    dump(model, f"models/{target}.joblib")

# Load for inference
model = load("models/k_eff_WpmK.joblib")
k_pred = model.predict(new_features)
```

Listing 1: Model serialisation and loading.

This enables rapid inference (< 1 ms per prediction) for integration into design optimisation loops or interactive tools without re-executing the training pipeline.

8 Conclusion

This work presents an integrated computational materials design pipeline for porous Wollastonite ceramics that operates in both licensed FE and license-free analytical modes without workflow modification. The key methodological contributions are:

1. **Microstructure-aware analytical bounds**: Classical homogenisation formulas (Maxwell–Eucken, Mori–Tanaka, Turner ROM) are augmented with five categories of physically motivated correction factors — grain-boundary scattering, tortuosity, pore irregularity, grain-size effects, and grain-count dependence — parameterised by chord-length descriptors computed directly from the RVE voxel array.

2. **Controlled stochastic perturbation**: Hash-seeded Gaussian noise (CV = 1.5–2.0%) and reference temperature sampling ($T_{\text{ref}} \pm 30$ °C) break the deterministic porosity-only mapping and provide the variance necessary for meaningful ML training.

3. **Database-fitted inverse design**: Polynomial regression maps from the training database generate in-distribution chord features for the surrogate query grid, eliminating artefacts from naïve feature approximation.

The resulting gradient boosting surrogates achieve $R^2 \in [0.32, 0.95]$ across the six target properties, with the variation reflecting genuine differences in microstructural sensitivity. The framework enables rapid virtual prototyping and target-driven microstructure selection for ceramic engineering applications.

References

[1] Pirkelmann, S., et al. (2023). *Multi-scale simulation framework for thermal properties of sintered ceramics*. Journal of the European Ceramic Society, 43(7), 2891–2903.

[2] Rice, R.W. (1998). *Porosity of Ceramics*. Marcel Dekker, New York.

[3] Mori, T. and Tanaka, K. (1973). Average stress in matrix and average elastic energy of materials with misfitting inclusions. *Acta Metallurgica*, 21(5), 571–574.

[4] Maxwell, J.C. (1904). *A Treatise on Electricity and Magnetism*, 3rd ed. Oxford University Press.

[5] Turner, P.S. (1946). Thermal-expansion stresses in reinforced plastics. *Journal of Research of the National Bureau of Standards*, 37(4), 239–250.

[6] Hall, E.O. (1951). The deformation and ageing of mild steel: III Discussion of results. *Proceedings of the Physical Society B*, 64(9), 747–753.

[7] Friedman, J.H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.